NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 2 EXAMINATION 2024-2025

EE6403 – DISTRIBUTED MULTIMEDIA SYSTEMS

April / May 2025 Time Allowed: 3 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 5 pages.
2. Answer all 4 questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.

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1. (a) Karhunen-Loeve Transform (KLT) is used to perform image compression for a class of images. The images are partitioned into 2×2 pixel blocks and ordered lexicographically to form 4×1 vectors. The covariance matrix of the vectors is given by:

$$\mathbf{C} = \begin{bmatrix} 4.0807 & 1.9915 & 5.6739 & 4.1269 \\ 1.9915 & 3.8484 & 0.7185 & 6.7208 \\ 5.6739 & 0.7185 & 10.8773 & 1.3651 \\ 4.1269 & 6.7208 & 1.3651 & 13.8936 \end{bmatrix}.$$

The covariance matrix $\mathbf{C}$ has four eigenvalues, λ_1, λ_2, λ_3 and λ_4 with corresponding eigenvectors $\mathbf{v}_1$, $\mathbf{v}_2$, $\mathbf{v}_3$ and $\mathbf{v}_4$. The eigenvalue λ_4 is known to have the smallest value amongst the four eigenvalues. The eigenvectors are orthogonal to each other.

Note: Question No. 1 continues on page 2.

(i) Find the eigenvalues λ_1, λ_2, and λ_3 given that

$$\mathbf{v}_1 = \begin{bmatrix} 0.3780 \\ 0.3780 \\ 0.3780 \\ 0.7559 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0.2965 \\ -0.2224 \\ 0.8154 \\ -0.4447 \end{bmatrix}, \quad \text{and } \mathbf{v}_3 = \begin{bmatrix} -0.2218 \\ 0.8875 \\ 0.1109 \\ -0.3883 \end{bmatrix}.$$

(ii) A 2×2 pixel block **A** after zero-mean centering is given by:

$$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}.$$

The pixel block **A** is to be compressed using KLT with the covariance matrix **C**. The user chooses to use the 2 most dominant principal components to compress **A**. Find the total square error between the original pixel block and the reconstructed pixel block.

(iii) The eigenvector $\mathbf{v}_4$ is given by:

$$\mathbf{v}_4 = \begin{bmatrix} 0.8486 \\ 0.1413 \\ a \\ b \end{bmatrix},$$

where a and b are real numbers. Find the values a and b. Round your answers to 3 decimal places.

(19 Marks)

(b) In a new KLT scheme, a user chooses to partition a new class of grayscale images into multiple 3×3 pixel blocks and then perform eigen-decomposition to construct the basis images. The user then uses the new KLT scheme to compress a typical grayscale image with size 30×30. Sketch the typical graph of total square error for the reconstructed image against the number of principal components used to represent the image.

(6 Marks)

2. (a) In an application, a user would like to compress a video using the MPEG-2 standard.

The video is to be displayed using the following format:

Resolution of the luminance plane: 720×480 pixels
Color depth of each pixel in each luminance and chrominance plane: 8 bits/pixel
Chroma subsampling format: 4:2:0
Frame rate: 30 frames per second

The Group-of-Pictures (GOP) structure of the video consists of one I-frame, p P-frames and q B-frames, where p and q are positive integers. The average compression ratios for the I-frames, P-frames, and B-frames are assumed to be 10:1, 20:1, and 40:1, respectively.

(i) Express and simplify the compression ratio of the compressed MPEG-2 video in terms of p and q. Write down the compression ratio if p=2 and q=6. State any assumption(s) you make in your calculation.

(ii) In a use case, a compressed video has a duration of 30 minutes, and it will be stored on Digital Versatile Disk (DVD). Assume that a GOP structure of IBBBPBBBPBBB is used. Find the storage requirement of the video in MB on the actual DVD (where 1 MB = 1024^2 bytes). State any assumption(s) you make in your calculation.

(17 Marks)

(b) Draw a diagram of the video structure / hierarchy for the MPEG-1 video bitstream. Clearly label all the key layers / levels in your diagram.

(8 Marks)

3. (a) A data sequence 1110001 (most significant bit on the left) is to be coded and transmitted using the Cyclic Redundancy Check (CRC) with the generator polynomial sequence of P = 1101.

(i) Determine the transmitted codeword.

(ii) Calculate the remainder obtained using the CRC at the receiver if 3 transmission errors occur in the third, fourth, and sixth bits (from the left) of the transmitted codeword obtained in part (i). Comment on the effectiveness of the CRC scheme in this case.

(12 Marks)

Note: Question No. 3 continues on page 4.

(b) An analog audio signal is given by $x(t) = 2m(t) + 3n(t)$, where $m(t)$ is a bandlimited signal occupying frequency band from 100 to 300 Hz, $n(t)$ is signal consisting of a single frequency at 200 Hz and t is given in second. A user would like to perform analog-to-digital conversion by converting the analog audio into digital audio such that it would not experience aliasing. Each sample would then be represented using 8 bits. Find the minimum bitrate of the digital audio in bits/s.

(6 Marks)

(c) With the aid of a diagram on Binary Frequency Shift Keying (BFSK), briefly explain 2 key differences between Amplitude Modulation (AM) and Frequency Shift Keying (FSK) in the modulation methods of communication systems.

(7 Marks)

4. (a) A Convolutional Neural Network (CNN) with the following structure is used to perform image classification.

Layer 1: Convolutional+RELU layer

Layer 2: Pooling layer

Layer 3: Convolutional+RELU layer

Layer 4: Pooling layer

Layer 5: Fully Connected (FC) layer

Layer 6: Softmax layer.

The FC layer of the CNN has the following values for a training sample:

Input to the FC layer: $\mathbf{x} = \begin{bmatrix}1 & 0 & 3 & 2\end{bmatrix}^{\mathrm{T}}$,

FC layer weight matrix: $\mathbf{W} = \begin{bmatrix} 0.1 & 0.1 & 0.1 & 0.1 \\ 0.4 & 0.4 & 0.2 & 0.1 \\ 0.2 & 0.6 & 0.5 & 0.4 \\ 0.3 & 0.1 & 0.3 & 0.1 \end{bmatrix}$,

FC layer bias: $\mathbf{b} = \begin{bmatrix}0.1 & 0.2 & -0.1 & -0.2\end{bmatrix}^{\mathrm{T}}$.

The ground truth output label of this training sample is given by $\mathbf{g} = \begin{bmatrix}0 & 1 & 0 & 0\end{bmatrix}^{\mathrm{T}}$.

Note: Question No. 4 continues on page 5.

(i) Find the output **p** after the Softmax layer.

(ii) Find the Softmax loss of this training sample.

(iii) Assume a 2×2 max pooling with a stride of 2 is used in Layer 4. Find the total number of trainable parameters in Layer 4, Layer 5 and Layer 6 of this CNN.

(17 Marks)

(b) A user would like to modify the structure of the CNN in part (a) to classify whether each of 5 independent clothing attributes (e.g., collar, sleeve, etc.) exists in a piece of clothing. Suggest <u>concisely</u> two main changes that should be made to the CNN structure in part (a) to address this problem.

(8 Marks)

END OF PAPER

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Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.